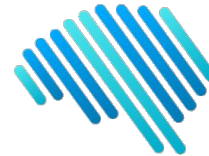




PONTIFICIA
UNIVERSIDAD
CATÓLICA
DE CHILE



IA Lab

Meta-aprendizaje (meta-learning)

Pablo Messina

Diplomado Inteligencia Artificial

Pontificia Universidad Católica de Chile

Tópicos de Profundización

Créditos

Algunas slides están basadas en el tutorial “Learning to learn: An Introduction to Meta Learning” por Chelsea Finn y Sergey Levine.

facebook research

The meta-learning problem

meta-learning:

$$\mathcal{D} = \{(x_1, y_1), \dots, (x_k, y_k)\}$$
$$\mathcal{D}_{\text{meta-train}} = \{\mathcal{D}_1, \dots, \mathcal{D}_n\}$$
$$\mathcal{D}_i = \{(x_1^i, y_1^i), \dots, (x_k^i, y_k^i)\}$$
$$\arg \max_{\phi} \log p(\phi | \mathcal{D}, \mathcal{D}_{\text{meta-train}})$$

what if we don't want to keep $\mathcal{D}_{\text{meta-train}}$ around forever?

learn meta-parameters θ : $p(\theta | \mathcal{D}_{\text{meta-train}})$

whatever we need to know about $\mathcal{D}_{\text{meta-train}}$ to solve new tasks

assume $\phi \perp\!\!\!\perp \mathcal{D}_{\text{meta-train}} | \theta$

$$\log p(\phi | \mathcal{D}, \mathcal{D}_{\text{meta-train}}) = \log \int_{\Theta} p(\phi | \mathcal{D}, \theta) p(\theta | \mathcal{D}_{\text{meta-train}}) d\theta$$
$$\approx \log p(\phi | \mathcal{D}, \theta^*) + \log p(\theta^* | \mathcal{D}_{\text{meta-train}})$$
$$\arg \max_{\phi} \log p(\phi | \mathcal{D}, \mathcal{D}_{\text{meta-train}}) \approx \arg \max_{\phi} \log p(\phi | \mathcal{D}, \theta^*)$$

this is the meta-learning problem

$$\theta^* = \arg \max_{\theta} \log p(\theta | \mathcal{D}_{\text{meta-train}})$$

Qs: slido.com/meta



Learning to learn: An Introduction to Meta Learning

Link a video:

<https://youtu.be/ByeRnmHJ-uk>

Link a slides:

<https://drive.google.com/file/d/1DuHyotdwEAEhmuHQQWwRosdiVBVGm8uYx/view>

Créditos

Algunas ideas y fórmulas fueron tomadas también del survey:

Meta-Learning in Neural Networks: A Survey

Timothy Hospedales, Antreas Antoniou, Paul Micaelli, Amos Storkey

Abstract—The field of meta-learning, or learning-to-learn, has seen a dramatic rise in interest in recent years. Contrary to conventional approaches to AI where tasks are solved from scratch using a fixed learning algorithm, meta-learning aims to improve the learning algorithm itself, given the experience of multiple learning episodes. This paradigm provides an opportunity to tackle many conventional challenges of deep learning, including data and computation bottlenecks, as well as generalization. This survey describes the contemporary meta-learning landscape. We first discuss definitions of meta-learning and position it with respect to related fields, such as transfer learning and hyperparameter optimization. We then propose a new taxonomy that provides a more comprehensive breakdown of the space of meta-learning methods today. We survey promising applications and successes of meta-learning such as few-shot learning and reinforcement learning. Finally, we discuss outstanding challenges and promising areas for future research.

Index Terms—Meta-Learning, Learning-to-Learn, Few-Shot Learning, Transfer Learning, Neural Architecture Search

Link: <https://arxiv.org/abs/2004.05439>

Paradigma tradicional: aprendizaje supervisado

Muchos datos
(+ modelos grandes)



Buena generalización

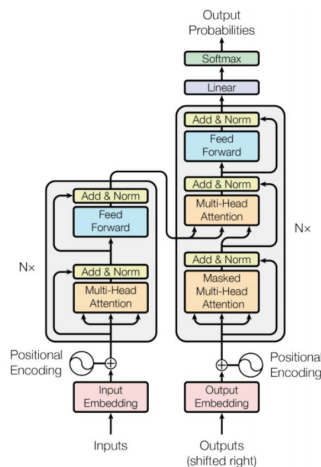
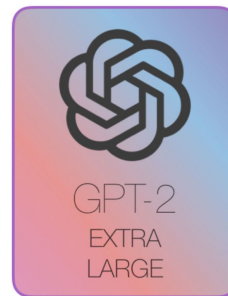


Figure 1: The Transformer - model architecture.



Problemas con el aprendizaje supervisado tradicional

- ¿Qué pasa si no tenemos un dataset grande?
 - Ej: imagenología médica, robótica, sistemas recomendadores, traducción de lenguajes poco comunes, etc.
- ¿Qué pasa si queremos una IA de propósito general en el mundo real?
 - Necesidad de adaptarse continuamente y aprender nuevas habilidades.
 - Aprender desde cero a cada rato sería súper ineficiente e impráctico → Necesidad de reutilizar conocimiento previo.
- ¿Qué pasa si los datos tienen una cola muy larga?



Ejemplo ...

training data

Braque



Cezanne



test datapoint



By Braque or Cezanne?

Ejemplo ...

training data

Braque



Cezanne



test datapoint



By Braque or Cezanne?

¿Cómo es que con 6 ejemplos podemos resolver esta tarea?

- Experiencia previa con muchas imágenes.
- Podemos adaptarnos a nuevas tareas rápido.

Ejemplo ...

training data

Braque



Cezanne



test datapoint



By Braque or Cezanne?

¿Cómo es que con 6 ejemplos podemos resolver esta tarea?

- Experiencia previa con muchas imágenes.
- Podemos adaptarnos a nuevas tareas rápido.

El Meta-aprendizaje nos permite resolver este problema

Agenda de la Clase

1. Meta-aprendizaje
 - a. Introducción simple
 - b. Formalizando algunos conceptos
 - c. Relación con Few-shot learning
2. Algunos algoritmos emblemáticos de meta-aprendizaje
 - a. Model-Agnostic Meta-Learning (MAML)
 - b. One-shot Learning with Memory-Augmented Neural Networks (MANN)
 - c. Métodos no-paramétricos
3. Aplicaciones
4. Trabajo práctico

Agenda de la Clase

1. Meta-aprendizaje

- a. Introducción simple
- b. Formalizando algunos conceptos
- c. Relación con Few-shot learning

2. Algunos algoritmos emblemáticos de meta-aprendizaje

- a. Model-Agnostic Meta-Learning (MAML)
- b. One-shot Learning with Memory-Augmented Neural Networks (MANN)
- c. Métodos no-paramétricos

3. Aplicaciones

4. Trabajo práctico

¿Qué es el Meta-aprendizaje?

- Objetivo: **aprender a aprender**
- Sirve cuando mi modelo va a enfrentar **tareas nuevas** (no vistas en entrenamiento), con relativamente **pocos ejemplos** por tarea, y por ende se necesita la capacidad de **aprender** o **adaptarse** eficientemente a nuevas tareas con pocos datos.
- O en términos más técnicos, sirve cuando el modelo debe aprender tanto en *train* como en *test*, donde el aprendizaje en test está limitado a pocos datos.
- IDEA CLAVE: Para aprender nuevas tareas eficientemente, el modelo debe *meta-aprender* una forma de *aprender* que sea eficiente, que le permita *aprender* nuevas tareas con pocos datos.

Meta-aprendizaje: explicación por analogía

Meta-aprendiendo cómo aprender un nuevo idioma: comparamos distintas estrategias de aprendizaje, las evaluamos, y nos quedamos con la estrategia que funcione mejor.

- Estrategia 1: aprender jugando con Duolingo.
- Estrategia 2: estudiar libros que enseñan el idioma (a la antigua).
- Estrategia 3: estudiar con ChatGPT.
- Estrategia 4: estudiar con Open English.
- Estrategia 5: hacer *shadowing* (*imitación*) de videos subtitrados en el idioma y consultar en un diccionario las palabras desconocidas.
- Etc.

Luego de evaluar estas estrategias, supongamos que descubrimos que la estrategia 5 es la mejor. Así, podemos decir que hemos *meta-aprendido* que la estrategia 5 es la mejor. Con este conocimiento, ahora podemos *aprender* un nuevo idioma, por ej., Chino mandarín utilizando la estrategia 5.

¿Qué cosas se pueden considerar como “estrategia de aprendizaje”?

En el contexto de machine learning, el concepto de “estrategia de aprendizaje” se puede interpretar de forma bastante amplia, pero puede incluir cosas como:

- Los parámetros con los que se inicializa el modelo (por ej. en el contexto de fine-tuning)
- Algún algoritmo genérico pre-aprendido que permite aprender nuevas clases anotando en una memoria
- Una estrategia genérica para aprender a realizar nuevas tareas por medio de instrucciones en lenguaje natural (por ej. los prompts en ChatGPT, Gemini, etc.)
- Qué hiper parámetros utilizar (arquitectura, optimizador, learning rate, función de pérdida, batch size, etc.)
- Etc.

Meta-aprendizaje: una definición un poco más formal

$\mathcal{D} = \{(x_1, y_1), \dots, (x_N, y_N)\}$, such as (input image, output label) pairs. We can train a predictive model $\hat{y} = f_{\theta}(x)$ parameterized by θ , by solving:

$$\theta^* = \arg \min_{\theta} \mathcal{L}(\mathcal{D}; \theta, \omega) \quad (1)$$

 Estrategia de aprendizaje

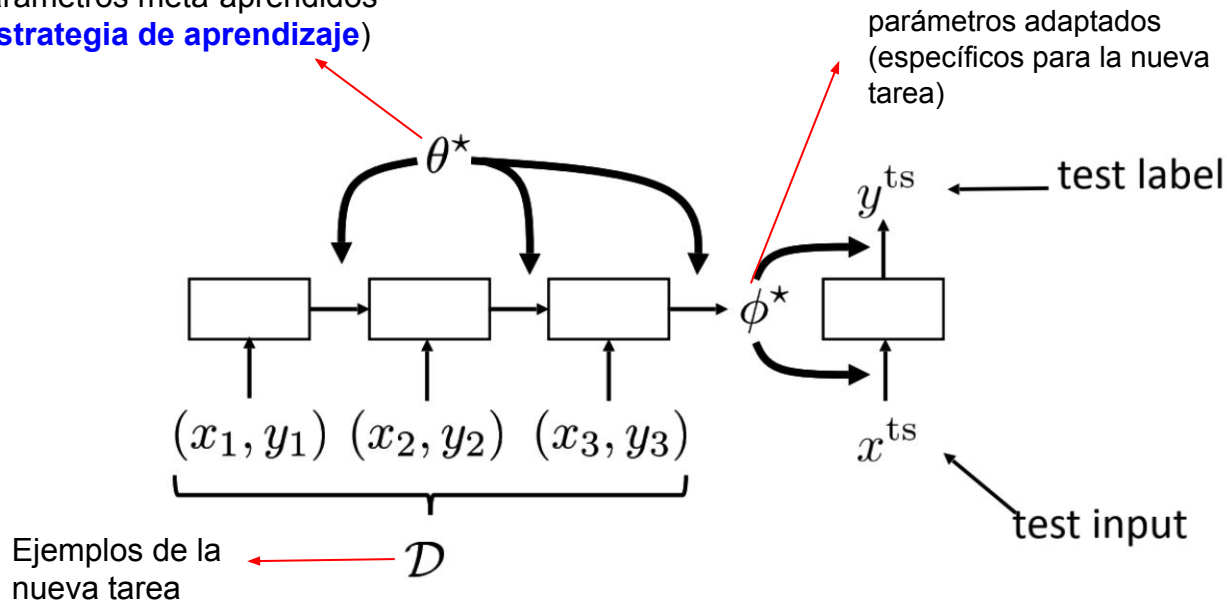
$$\omega^* = \arg \min_{\omega} \sum_{i=1}^M \mathcal{L}^{meta}(\theta^{*(i)}(\omega), \omega, \mathcal{D}_{source}^{val(i)}) \quad (5)$$

$$\text{s.t. } \theta^{*(i)}(\omega) = \arg \min_{\theta} \mathcal{L}^{task}(\theta, \omega, \mathcal{D}_{source}^{train(i)}) \quad (6)$$

$$\mathcal{D}_{source} = \{(\mathcal{D}_{source}^{train}, \mathcal{D}_{source}^{val})^{(i)}\}_{i=1}^M \quad \mathcal{D}_{target} = \{(\mathcal{D}_{target}^{train}, \mathcal{D}_{target}^{test})^{(i)}\}_{i=1}^Q$$

Meta-aprendizaje: una explicación visual

parámetros meta-aprendidos
(**estrategia de aprendizaje**)



Meta-training y Meta-testing

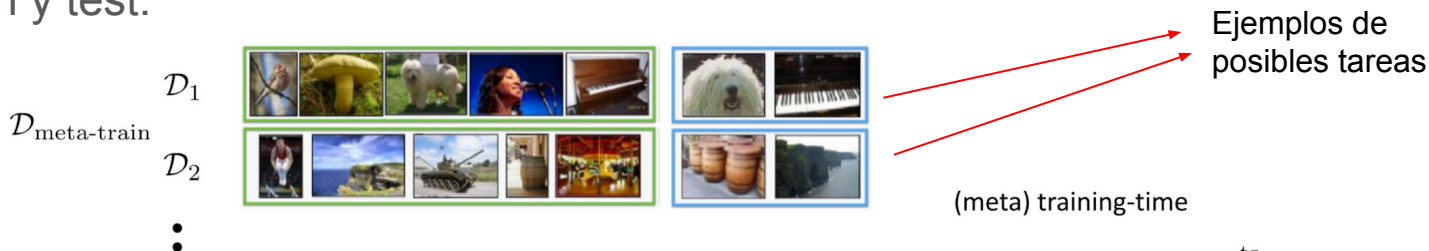
El meta-aprendizaje contempla dos etapas:

Meta-training: los parámetros del modelo se optimizan para **aprender la habilidad de adaptarse** a nuevas tareas.

Meta-testing: el modelo **usa su habilidad de adaptación** (ya aprendida) sobre nuevas tareas.

Meta-training

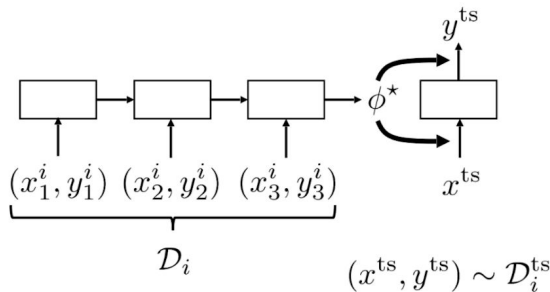
- Se busca aprender la habilidad de adaptación (parámetros θ).
- Cada instancia de meta-entrenamiento representa una tarea diferente con ejemplos de train y test.



$$\mathcal{D}_{\text{meta-train}} = \{(\mathcal{D}_1^{\text{tr}}, \mathcal{D}_1^{\text{ts}}), \dots, (\mathcal{D}_n^{\text{tr}}, \mathcal{D}_n^{\text{ts}})\}$$

$$\mathcal{D}_i^{\text{tr}} = \{(x_1^i, y_1^i), \dots, (x_k^i, y_k^i)\}$$

$$\mathcal{D}_i^{\text{ts}} = \{(x_1^i, y_1^i), \dots, (x_l^i, y_l^i)\}$$



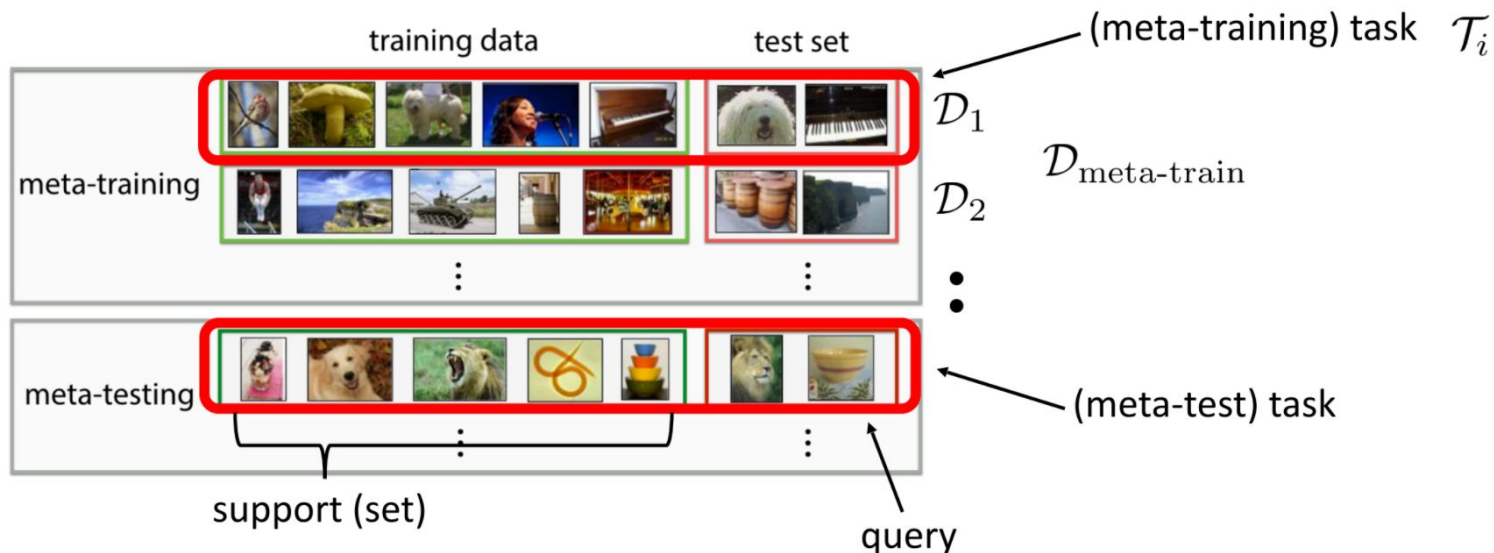
Key idea:

“our training procedure is based on a simple machine learning principle: test and train conditions must match”

Vinyals et al., Matching Networks for One-Shot Learning

Meta-testing

- Se busca evaluar la habilidad de adaptación (parámetros Φ).
- Cada instancia de meta-testing representa una tarea diferente con ejemplos de train y test (igual que en en meta-training).



Relación con Few-shot learning

- Few-shot learning tiene que ver con el planteamiento del problema (N-way, K-shot)
- Meta-aprendizaje tiene que ver con la estrategia de aprendizaje (aplicable a few-shot learning)

Training task 1

Support set

K=2



N=3

Query set



Training task 2 . . .

Support set

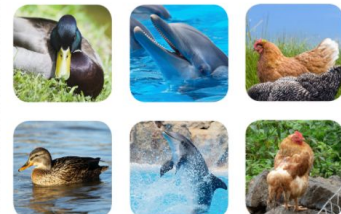


Query set



Test task 1 . . .

Support set



Query set



Datasets de Few-shot learning

- Omniglot: <https://github.com/brendenlake/omniglot>

the Omniglot dataset Lake et al. Science 2015

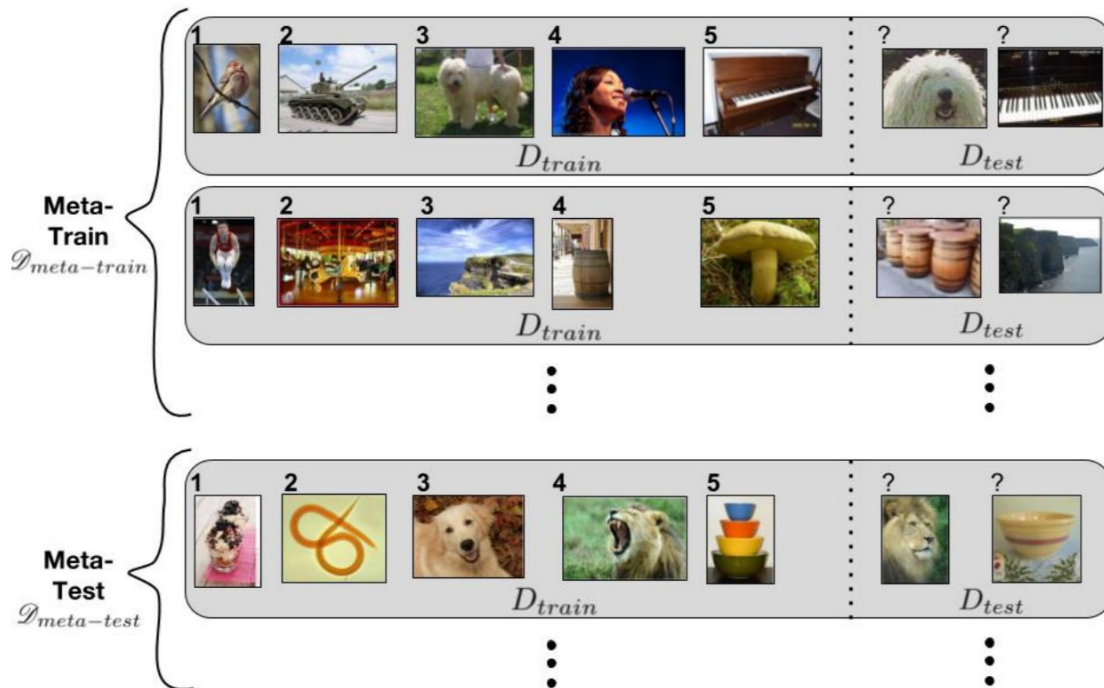
1623 characters from 50 different alphabets



20 instances of each character

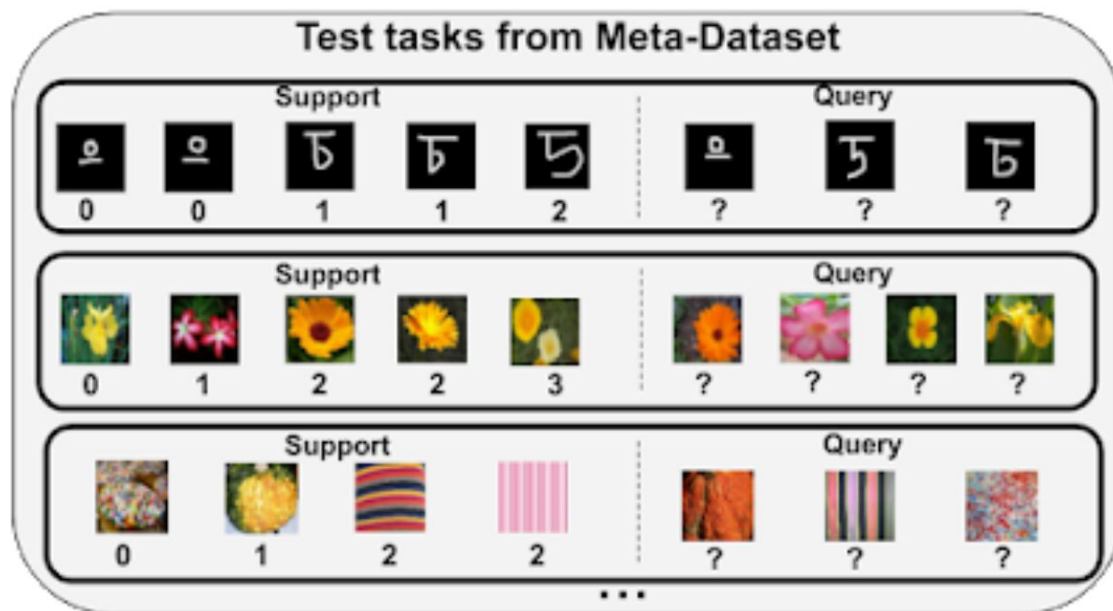
Datasets de Few-shot learning

- Mini-imagenet: <https://github.com/yaoyao-liu/mini-imagenet-tools>



Datasets de Few-shot learning

- Meta-Dataset: <https://ai.googleblog.com/2020/05/announcing-meta-dataset-dataset-of.html>



Agenda de la Clase

1. Meta-aprendizaje

- a. Introducción simple
- b. Formalizando algunos conceptos
- c. Relación con Few-shot learning

2. Algunos algoritmos emblemáticos de meta-aprendizaje

- a. Model-Agnostic Meta-Learning (MAML)
- b. One-shot Learning with Memory-Augmented Neural Networks (MANN)
- c. Métodos no-paramétricos

3. Aplicaciones

4. Trabajo práctico

MAML (2017)

Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks

Chelsea Finn¹ Pieter Abbeel^{1,2} Sergey Levine¹

Link a paper: <https://arxiv.org/abs/1703.03400>

MAML (2017)

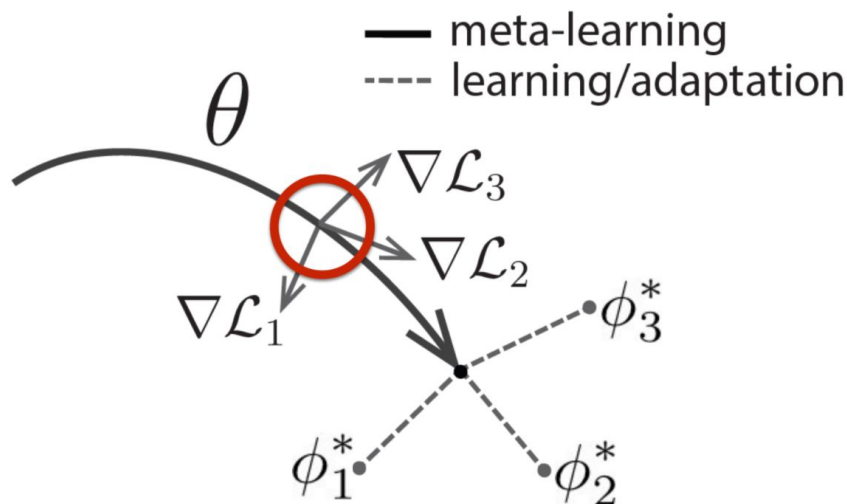
- Se propone un **algoritmo** de meta-entrenamiento para encontrar una **buena inicialización** de los parámetros de un modelo.
- En meta-test, el modelo ajusta sus parámetros a la nueva tarea por medio de un proceso de **fine-tuning** más eficiente.
- El algoritmo es **model agnostic**, es decir, se puede aplicar a cualquier modelo.

MAML (2017)

$$\min_{\theta} \sum_{\text{task } i} \mathcal{L}(\theta - \alpha \nabla_{\theta} \mathcal{L}(\theta, \mathcal{D}_i^{\text{tr}}), \mathcal{D}_i^{\text{ts}})$$

θ parameter vector
being meta-learned

ϕ_i^* optimal parameter
vector for task i



MAML (2017)

Fine-tuning
[test-time]

$$\phi \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}(\theta, \mathcal{D}^{\text{tr}})$$

pre-trained parameters

training data for new task

Meta-learning

$$\min_{\theta} \sum_{\text{task } i} \mathcal{L}(\theta - \alpha \nabla_{\theta} \mathcal{L}(\theta, \mathcal{D}_i^{\text{tr}}), \mathcal{D}_i^{\text{ts}})$$

Key idea: Over many tasks, learn parameter vector θ that transfers via fine-tuning

MAML (2017)

Idea clave: obtener ϕ_i a través de un proceso de **optimización**.

General Algorithm:

Optimization-based approach

1. Sample task $\mathcal{T}_i$ (*or mini batch of tasks*)
 2. Sample disjoint datasets $\mathcal{D}_i^{\text{tr}}, \mathcal{D}_i^{\text{test}}$ from $\mathcal{D}_i$
 3. Optimize $\phi_i \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}(\theta, \mathcal{D}_i^{\text{tr}})$
 4. Update θ using $\nabla_{\theta} \mathcal{L}(\phi_i, \mathcal{D}_i^{\text{test}})$
- 

—> brings up **second-order** derivatives

MAML (2017)

Algunos resultados:

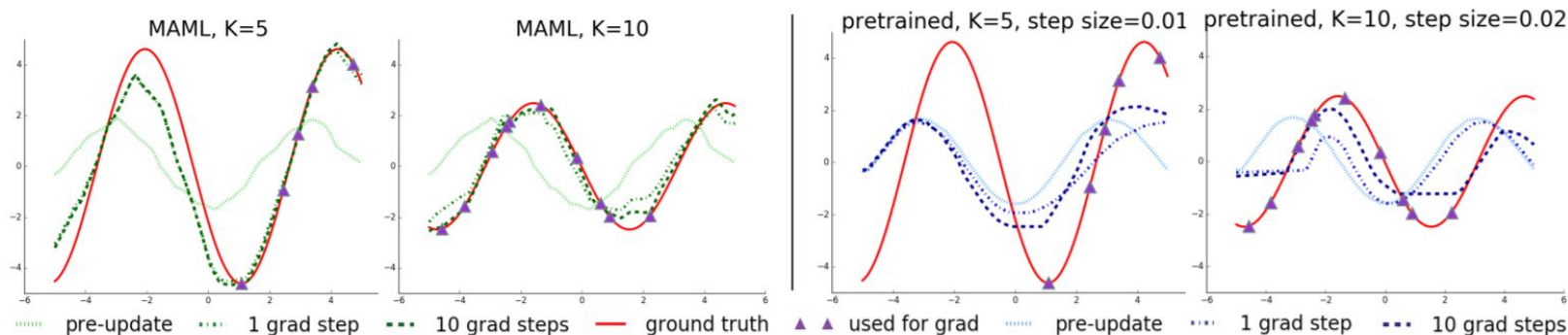


Figure 2. Few-shot adaptation for the simple regression task. Left: Note that MAML is able to estimate parts of the curve where there are no datapoints, indicating that the model has learned about the periodic structure of sine waves. Right: Fine-tuning of a model pretrained on the same distribution of tasks without MAML, with a tuned step size. Due to the often contradictory outputs on the pre-training tasks, this model is unable to recover a suitable representation and fails to extrapolate from the small number of test-time samples.

MAML (2017)

Algunos resultados:

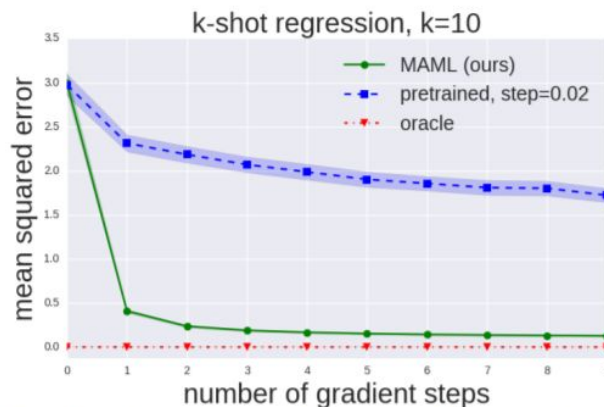


Figure 3. Quantitative sinusoid regression results showing the learning curve at meta test-time. Note that MAML continues to improve with additional gradient steps without overfitting to the extremely small dataset during meta-testing, achieving a loss that is substantially lower than the baseline fine-tuning approach.

MAML (2017)

Algunos resultados:

Table 1. Few-shot classification on held-out Omniglot characters (top) and the MiniImagenet test set (bottom). MAML achieves results that are comparable to or outperform state-of-the-art convolutional and recurrent models. Siamese nets, matching nets, and the memory module approaches are all specific to classification, and are not directly applicable to regression or RL scenarios. The $\pm$ shows 95% confidence intervals over tasks. Note that the Omniglot results may not be strictly comparable since the train/test splits used in the prior work were not available. The MiniImagenet evaluation of baseline methods and matching networks is from Ravi & Larochelle (2017).

Omniglot (Lake et al., 2011)	5-way Accuracy		20-way Accuracy	
	1-shot	5-shot	1-shot	5-shot
MANN, no conv (Santoro et al., 2016)	82.8%	94.9%	–	–
MAML, no conv (ours)	$89.7 \pm 1.1\%$	$97.5 \pm 0.6\%$	–	–
Siamese nets (Koch, 2015)	97.3%	98.4%	88.2%	97.0%
matching nets (Vinyals et al., 2016)	98.1%	98.9%	93.8%	98.5%
neural statistician (Edwards & Storkey, 2017)	98.1%	99.5%	93.2%	98.1%
memory mod. (Kaiser et al., 2017)	98.4%	99.6%	95.0%	98.6%
MAML (ours)	$98.7 \pm 0.4\%$	$99.9 \pm 0.1\%$	$95.8 \pm 0.3\%$	$98.9 \pm 0.2\%$

MiniImagenet (Ravi & Larochelle, 2017)	5-way Accuracy	
	1-shot	5-shot
fine-tuning baseline	$28.86 \pm 0.54\%$	$49.79 \pm 0.79\%$
nearest neighbor baseline	$41.08 \pm 0.70\%$	$51.04 \pm 0.65\%$
matching nets (Vinyals et al., 2016)	$43.56 \pm 0.84\%$	$55.31 \pm 0.73\%$
meta-learner LSTM (Ravi & Larochelle, 2017)	$43.44 \pm 0.77\%$	$60.60 \pm 0.71\%$
MAML, first order approx. (ours)	$48.07 \pm 1.75\%$	$63.15 \pm 0.91\%$
MAML (ours)	$48.70 \pm 1.84\%$	$63.11 \pm 0.92\%$

MAML (2017)

Algunos resultados:

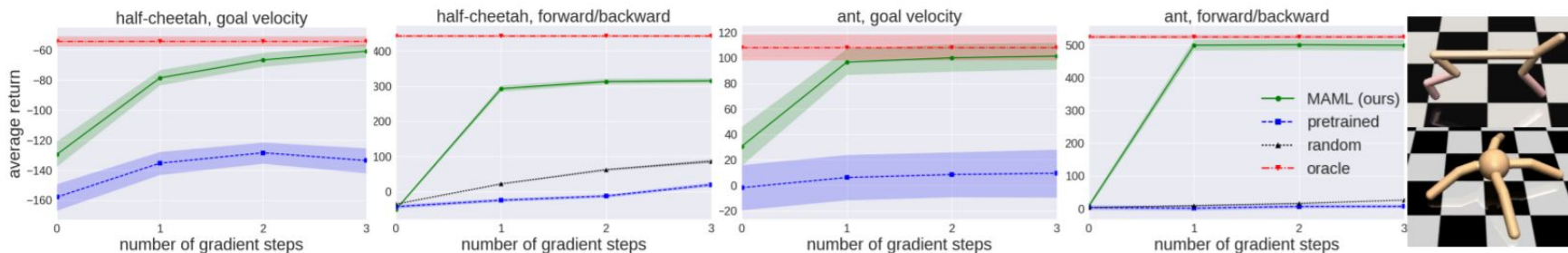


Figure 5. Reinforcement learning results for the half-cheetah and ant locomotion tasks, with the tasks shown on the far right. Each gradient step requires additional samples from the environment, unlike the supervised learning tasks. The results show that MAML can adapt to new goal velocities and directions substantially faster than conventional pretraining or random initialization, achieving good performs in just two or three gradient steps. We exclude the goal velocity, random baseline curves, since the returns are much worse (< -200 for cheetah and < -25 for ant).

One-shot Learning with Memory-Augmented Neural Networks (2016)

One-shot Learning with Memory-Augmented Neural Networks

Adam Santoro

Google DeepMind

ADAMSANTORO@GOOGLE.COM

Sergey Bartunov

Google DeepMind, National Research University Higher School of Economics (HSE)

SBOS@SBOS.IN

Matthew Botvinick

Daan Wierstra

Timothy Lillicrap

Google DeepMind

BOTVINICK@GOOGLE.COM

WIERSTRA@GOOGLE.COM

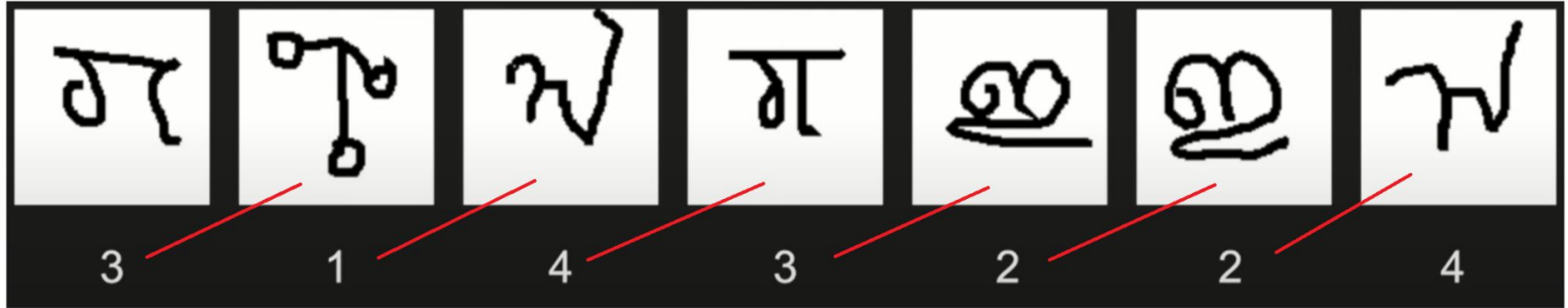
COUNTZERO@GOOGLE.COM

Link a paper: <https://arxiv.org/abs/1605.06065>

One-shot Learning with Memory-Augmented Neural Networks (2016)

- Se propone un **modelo aumentado con memoria** con el objetivo de aprender a reconocer y etiquetar nuevos caracteres (combinaciones nunca vistas en entrenamiento) a partir de **un solo ejemplo** (one-shot learning).
- Se propone además un **algoritmo** de meta-entrenamiento que fuerza al modelo a aprender a usar la memoria.
- En meta-test, los parámetros se mantienen congelados (no hay fine-tuning).

One-shot Learning with Memory-Augmented Neural Networks (2016)



Para interesados, este video lo explica bastante bien: <https://youtu.be/KUWYwwwvQv8E?t=490>

One-shot Learning with Memory-Augmented Neural Networks (2016)

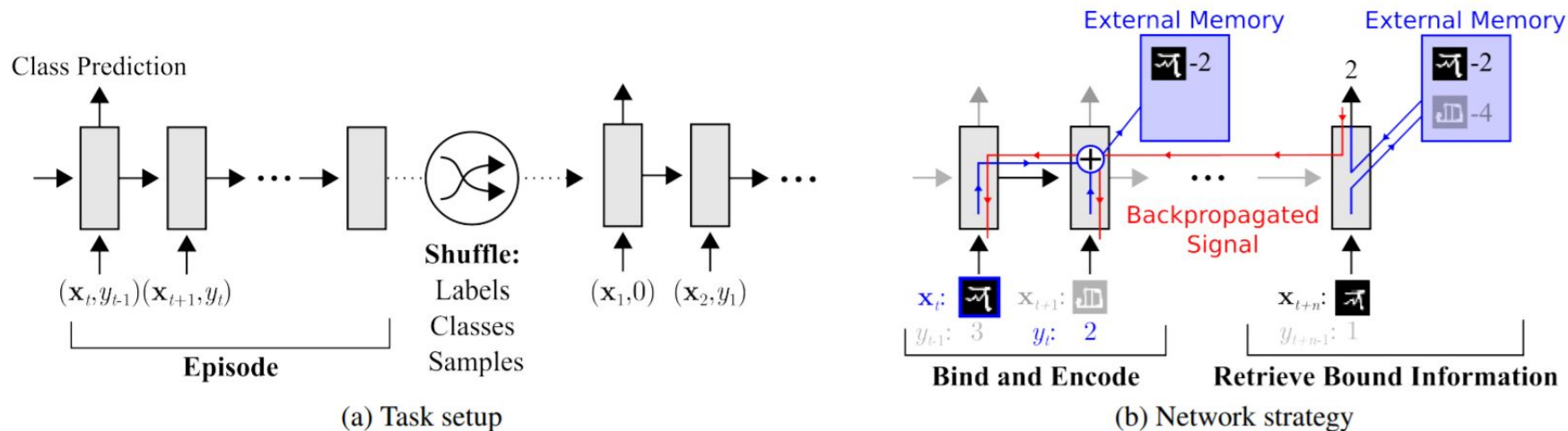
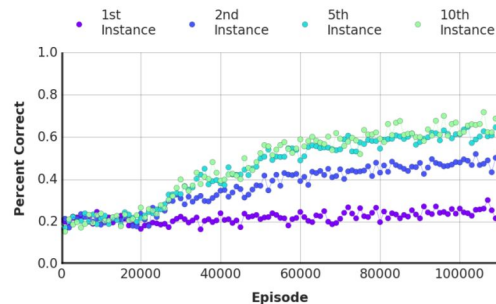


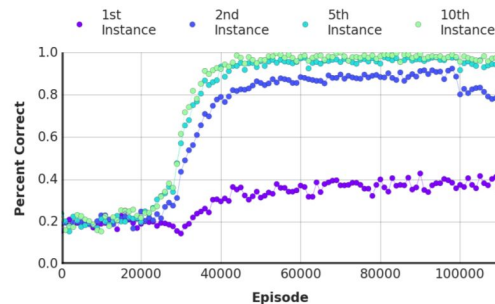
Figure 1. Task structure. (a) Omniglot images (or x -values for regression), x_t , are presented with time-offset labels (or function values), y_{t-1} , to prevent the network from simply mapping the class labels to the output. From episode to episode, the classes to be presented in the episode, their associated labels, and the specific samples are all shuffled. (b) A successful strategy would involve the use of an external memory to store bound sample representation-class label information, which can then be retrieved at a later point for successful classification when a sample from an already-seen class is presented. Specifically, sample data x_t from a particular time step should be bound to the appropriate class label y_t , which is presented in the subsequent time step. Later, when a sample from this same class is seen, it should retrieve this bound information from the external memory to make a prediction. Backpropagated error signals from this prediction step will then shape the weight updates from the earlier steps in order to promote this binding strategy.

One-shot Learning with Memory-Augmented Neural Networks (2016)

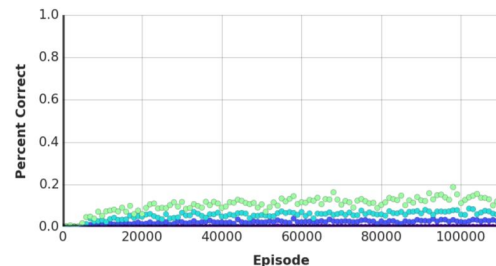
Algunos resultados:



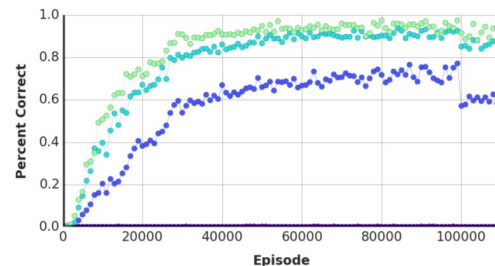
(a) LSTM, five random classes/episode, one-hot vector labels



(b) MANN, five random classes/episode, one-hot vector labels



(c) LSTM, fifteen classes/episode, five-character string labels



(d) MANN, fifteen classes/episode, five-character string labels

Figure 2. Omniglot classification. The network was given either five (a-b) or up to fifteen (c-d) random classes per episode, which were of length 50 or 100 respectively. Labels were one-hot vectors in (a-b), and five-character strings in (c-d). In (b), first instance accuracy is above chance, indicating that the MANN is performing “educated guesses” for new classes based on the classes it has already seen and stored in memory. In (c-d), first instance accuracy is poor, as is expected, since it must make a guess from 3125 random strings. Second instance accuracy, however, approaches 80% during training for the MANN (d). At the 100,000 episode mark the network was tested, without further learning, on distinct classes withheld from the training set, and exhibited comparable performance.

One-shot Learning with Memory-Augmented Neural Networks (2016)

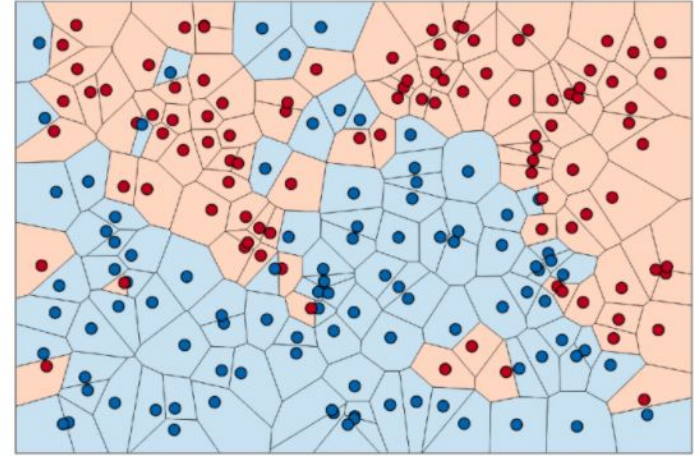
Algunos
resultados:

Table 1. Test-set classification accuracies for humans compared to machine algorithms trained on the Omniglot dataset, using one-hot encodings of labels and five classes presented per episode.

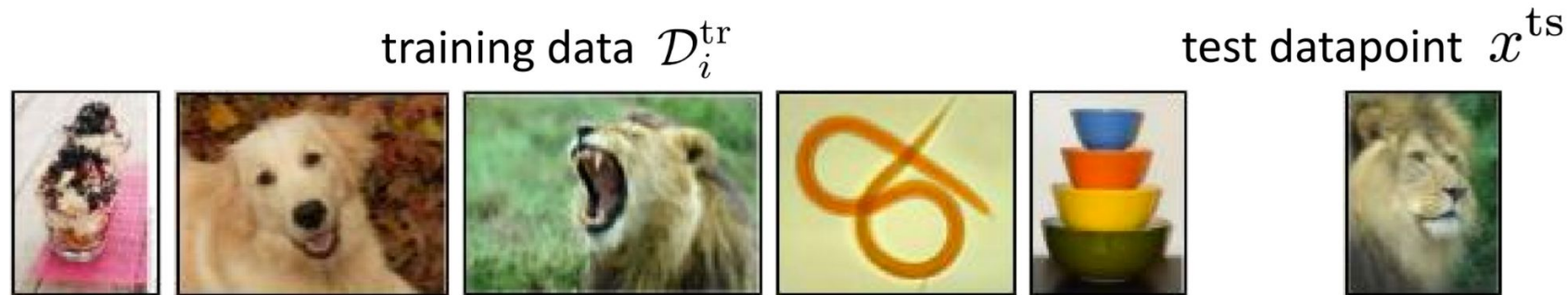
MODEL	INSTANCE (% CORRECT)					
	1 ST	2 ND	3 RD	4 TH	5 TH	10 TH
HUMAN	34.5	57.3	70.1	71.8	81.4	92.4
FEEDFORWARD	24.4	19.6	21.1	19.9	22.8	19.5
LSTM	24.4	49.5	55.3	61.0	63.6	62.5
MANN	36.4	82.8	91.0	92.6	94.9	98.1

Métodos no-paramétricos

- Buscan resolver **tareas de clasificación** con una **cantidad potencialmente ilimitada** de clases.
- Se aborda el problema a través de el **aprendizaje de una métrica de distancia** que permita asignar vectores similares a objetos similares (algo así como un “KNN meta-aprendido”)



Métodos no-paramétricos



Compare test image with training images

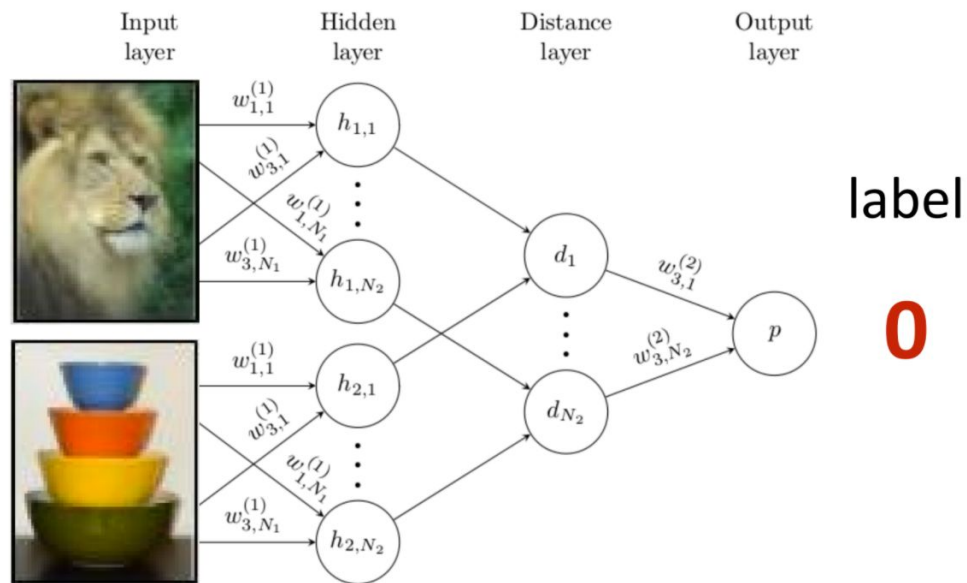
In what space do you compare? With what distance metric?

pixel space, l_2 -distance?

Learn to compare using data!

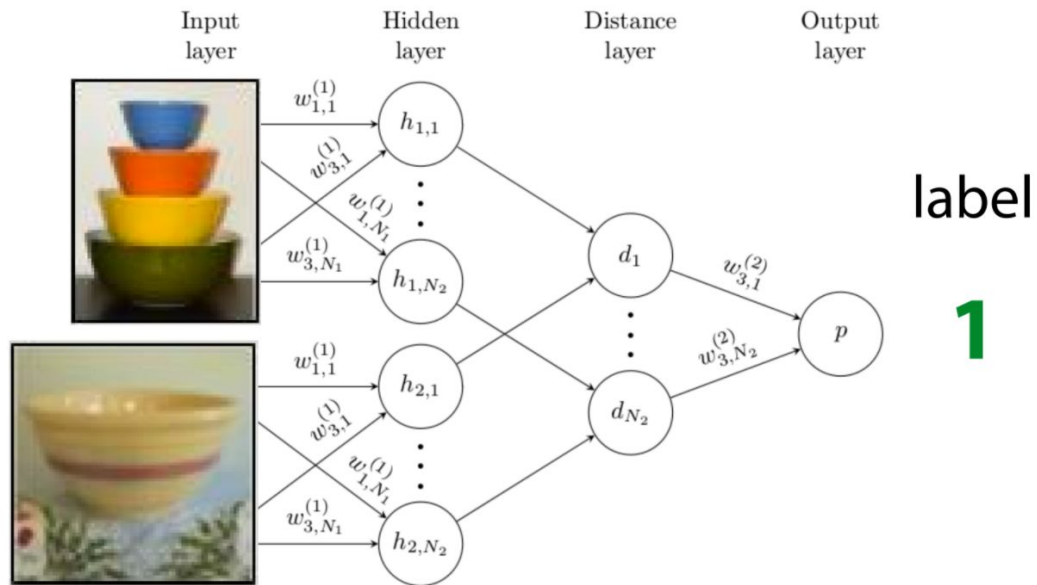
Métodos no-paramétricos

1) Redes Siamesas



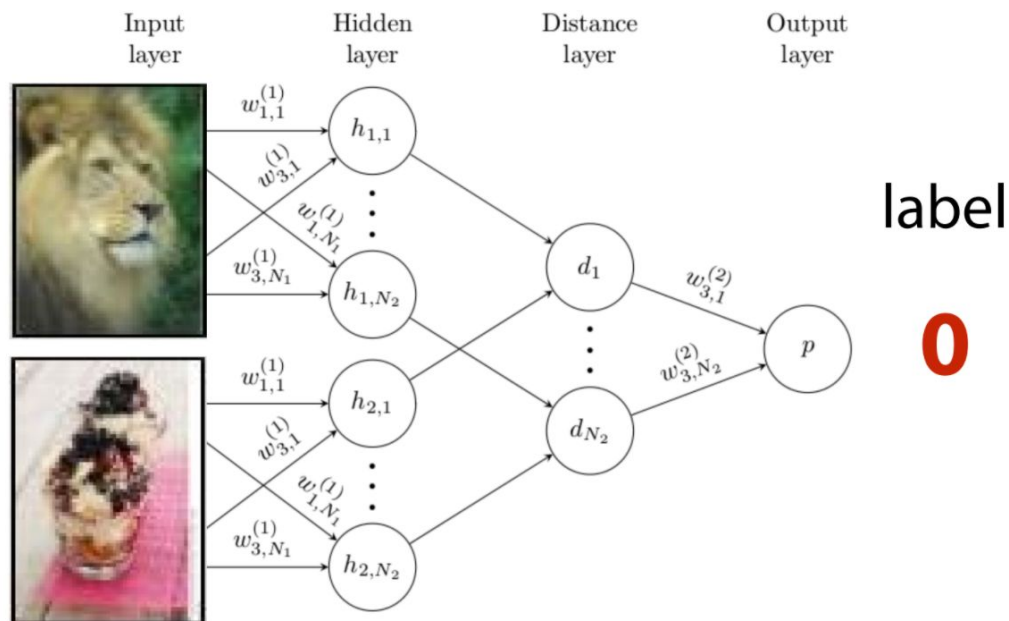
Métodos no-paramétricos

1) Redes Siamesas



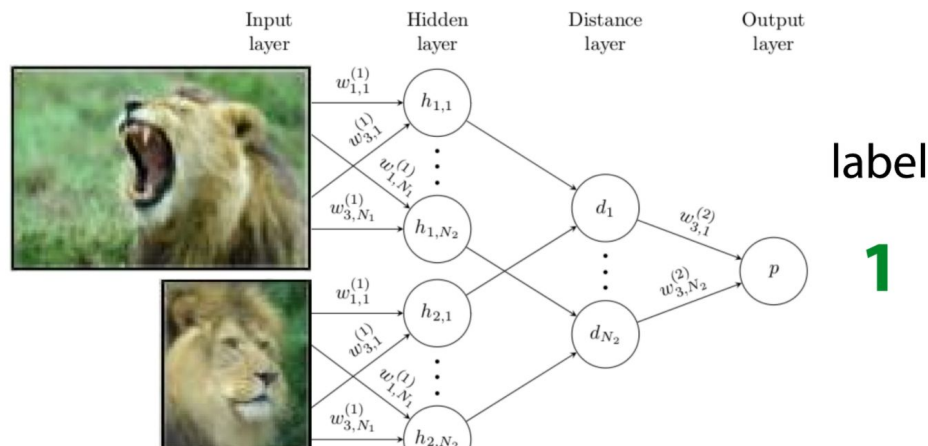
Métodos no-paramétricos

1) Redes Siamesas



Métodos no-paramétricos

1) Redes Siamesas



Meta-test time: compare image $\mathbf{x}_{\text{test}}$ to each image in $\mathcal{D}_j^{\text{tr}}$

Meta-training: 2-way classification

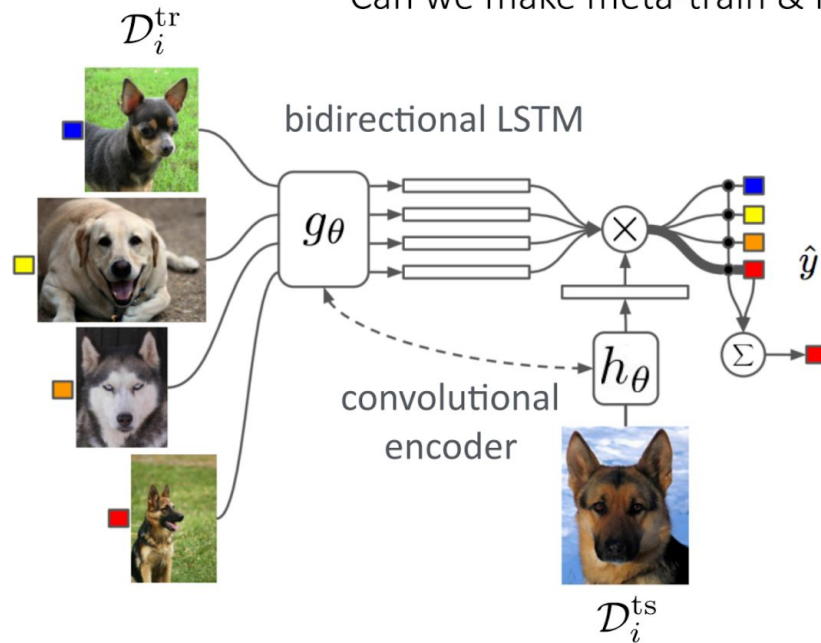
Meta-test: N-way classification

Can we **match** meta-train & meta-test?

Métodos no-paramétricos

2) Matching Networks

Can we make meta-train & meta-test match?



Weighed nearest neighbors in
learned embedding space

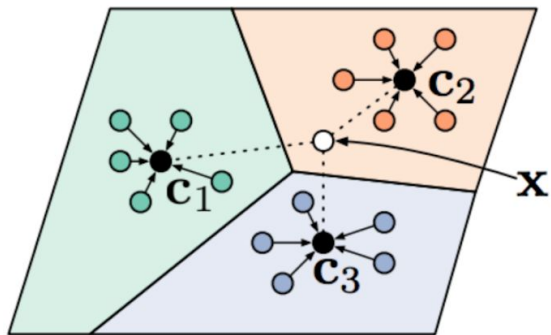
$$\hat{y} = \sum_{i=1}^k a(\hat{x}, x_i) y_i$$

What if >1 shot?

Can we aggregate class information to
create a **prototypical embedding**?

Métodos no-paramétricos

3) Prototypical Networks



(a) Few-shot

$$\mathbf{c}_k = \frac{1}{|\mathcal{D}_i^{\text{tr}}|} \sum_{(x,y) \in \mathcal{D}_i^{\text{tr}}} f_{\theta}(x)$$

$$p_{\theta}(y = k|x) = \frac{\exp(-d(f_{\theta}(x), \mathbf{c}_k))}{\sum_{k'} \exp(-d(f_{\theta}(x), \mathbf{c}_{k'}))}$$

d: Euclidean, or cosine distance

Agenda de la Clase

1. Meta-aprendizaje

- a. Introducción simple
- b. Formalizando algunos conceptos
- c. Relación con Few-shot learning

2. Algunos algoritmos emblemáticos de meta-aprendizaje

- a. Model-Agnostic Meta-Learning (MAML)
- b. One-shot Learning with Memory-Augmented Neural Networks (MANN)
- c. Métodos no-paramétricos

3. Aplicaciones

4. Trabajo práctico

Aplicaciones en medicina

Semi-supervised Meta-learning with Disentanglement for Domain-generalised Medical Image Segmentation

Xiao Liu¹, Spyridon Thermos¹, Alison O'Neil^{1,3} and Sotirios A. Tsaftaris^{1,2}

¹ School of Engineering, University of Edinburgh, Edinburgh EH9 3FB, UK

² The Alan Turing Institute, London, UK

³ Canon Medical Research Europe Ltd., Edinburgh, UK

Xiao.Liu@ed.ac.uk, SThermos@ed.ac.uk, Alison.ONeil@mre.medical.canon,
S.Tsaftaris@ed.ac.uk

Link a paper: <https://arxiv.org/abs/2106.13292> (MICCAI 2021 Oral)

Aplicaciones en medicina

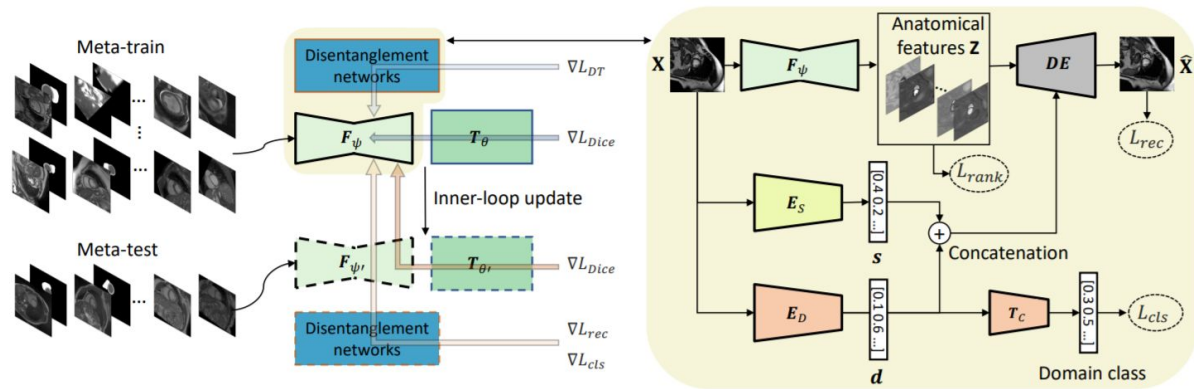


Fig. 1. At each iteration, the training dataset is split into meta-train and meta-test sets including labeled and unlabeled data. A feature network F_ψ extracts features Z for a task network T_θ to predict segmentation masks. The model is trained in a semi-supervised setting, where $\mathcal{L}_{DT}$, $\mathcal{L}_{rec}$ and $\mathcal{L}_{cls}$ do not require pixel-wise annotation. In the inner-loop update, ψ' and θ' are computed for the meta-test step (see Eq. 1). Finally, all the gradients are computed to update F_ψ and T_θ as in Eq. 2. The disentanglement networks decompose image X to common s and specific to the domain d representations to be disentangled with Z for meta-train and meta-test sets with the constraints ($\mathcal{L}_{DT}$ and $\mathcal{L}_{rec}$ and $\mathcal{L}_{cls}$). See Section 2 for loss definitions.

Aplicaciones en medicina

Papers con meta-learning en MICCAI 2025: <https://papers.miccai.org/miccai-2025/>

- **Fit Pixels, Get Labels: Meta-Learned Implicit Networks for Image Segmentation (BEST PAPER AWARD)**
Vyas, Kushal; Veeraraghavan, Ashok; Balakrishnan, Guha;
[PDF] [Paper Information and Reviews] [bibtex]
- **Meta-Learning Physics-Informed Neural Networks for Personalized Cardiac Modeling**
Toloubidokhti, Maryam; Missel, Ryan; Lian, Shichang; Wang, Linwei;
[PDF] [Paper Information and Reviews] [bibtex]
- **Meta-Learning-Driven CT Morphology Disentangled Diffusion Model for Multi-Region SPECT Attenuation Correction**
Yang, Haoran; Fan, Jiansong; Li, Lihua; Pan, Xiang;
[PDF] [Paper Information and Reviews] [bibtex]
- **Sparsely Labeled fMRI Data Denoising with Meta-Learning-Based Semi-Supervised Domain Adaptation**
Heo, Keun-Soo; Han, Ji-Wung; Bak, Soyeon; Lim, Minjoo; Kang, Bogyeong; Park, Sang-Jun; Lin, Weili; Zhang, Han; Shen, Dinggang; Kam, Tae-Eui;
[PDF] [Paper Information and Reviews] [bibtex]

Aplicaciones en medicina



This MICCAI paper is the Open Access version, provided by the MICCAI Society. It is identical to the accepted version, except for the format and this watermark; the final published version is available on SpringerLink.

Fit Pixels, Get Labels: Meta-Learned Implicit Networks for Image Segmentation

Kushal Vyas^[0009-0000-5952-697X], Ashok Veeraraghavan^[0000-0001-5043-7460],
and Guha Balakrishnan^[0000-0001-8703-1368]

Rice University, Houston TX 77005, USA
guha@rice.edu

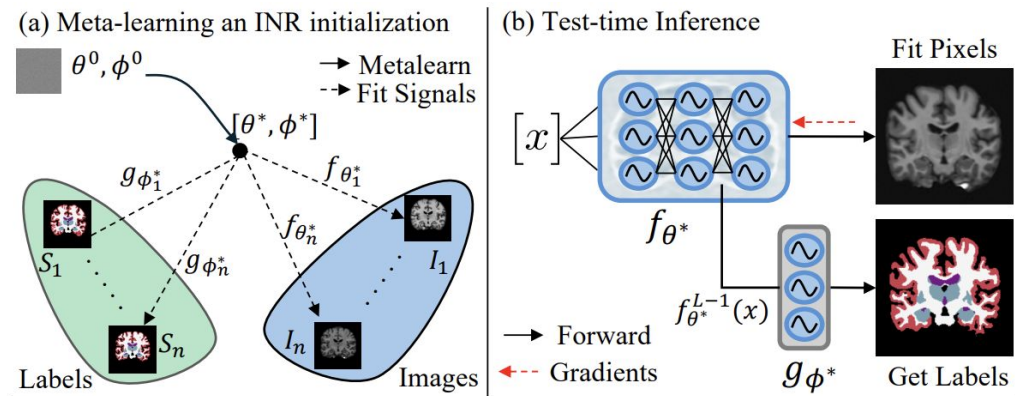


Fig. 1. Overview of MetaSeg. (a) We use a meta-learning framework to learn optimal initial parameters θ^*, ϕ^* for an INR consisting of an L -layer reconstruction network $f_{\theta}(\cdot)$ and shallow segmentation head $g_{\phi}(\cdot)$. (b) At test time, optimally initialized INR f_{θ^*} is iteratively fit to the pixels of an unseen test scan. After convergence, the penultimate features $f_{\theta^*}^{L-1}(x)$ are fed as input to the segmentation head $g_{\phi^*}(\cdot)$ to predict per-pixel class labels.

Aplicaciones en medicina



This MICCAI paper is the Open Access version, provided by the MICCAI Society. It is identical to the accepted version, except for the format and this watermark; the final published version is available on SpringerLink.

Sparsely Labeled fMRI Data Denoising with Meta-Learning-Based Semi-Supervised Domain Adaptation

Keun-Soo Heo¹, Ji-Wung Han¹, Soyeon Bak¹, Minjoo Lim¹, Bogyong Kang¹, Sang-Jun Park¹, Weili Lin², Han Zhang³, Dinggang Shen³, and Tae-Eui Kam^{1*}

¹ Department of Artificial Intelligence, Korea University, Republic of Korea
kamte@korea.ac.kr

² Department of Radiology, University of North Carolina at Chapel Hill, USA

³ School of Biomedical Engineering, ShanghaiTech University, China

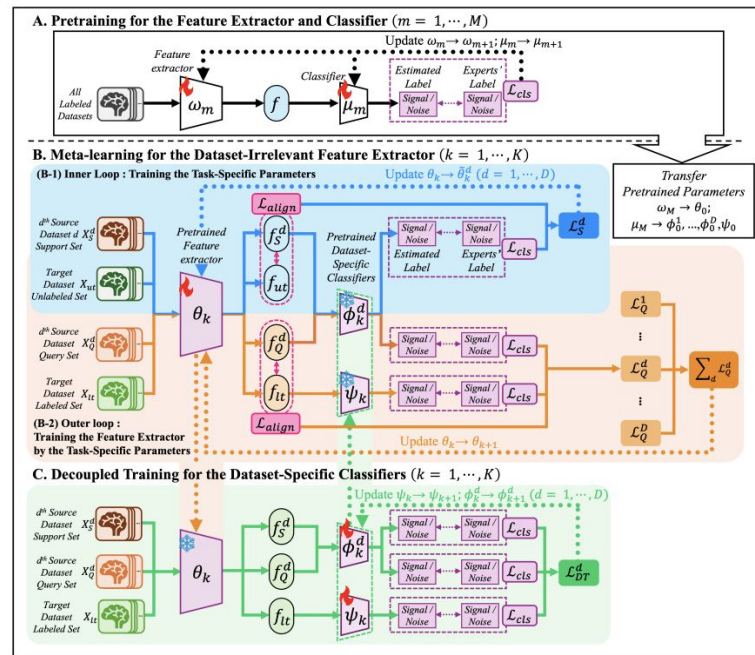


Fig. 1. Overview of our proposed framework for fMRI denoising, which learns dataset-irrelevant features from a sparsely labeled dataset (a target domain) by leveraging existing labeled datasets (source domains) with three stages: (A) pretraining for the feature extractor and classifier, (B) meta-learning for the dataset-irrelevant feature extractor, and (C) decoupled training for the dataset-specific classifiers.

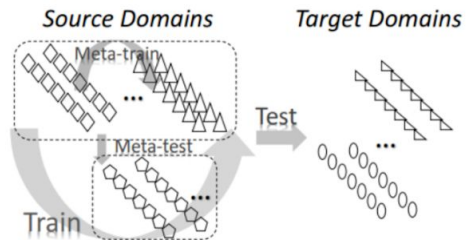
Aplicaciones en computer vision

few-shot image recognition



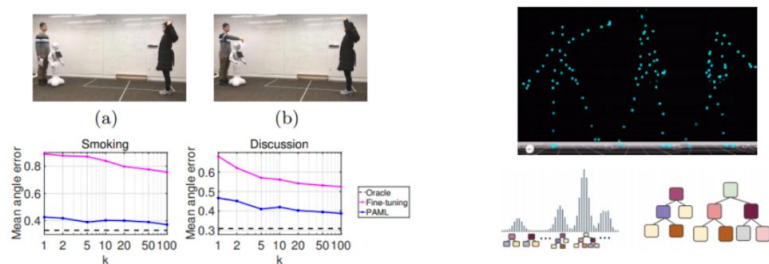
see, e.g.: Vinyals et al. **Matching Networks for One Shot Learning**, and many many others

domain adaptation



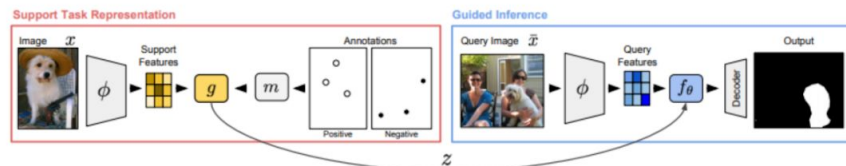
see, e.g.: Li, Yang, Song, Hospedales. **Learning to Generalize: Meta-Learning for Domain Adaptation**.

human motion and pose prediction



see, e.g.: Gui et al. **Few-Shot Human Motion Prediction via Meta-Learning**.
Alet et al. **Modular Meta-Learning**.

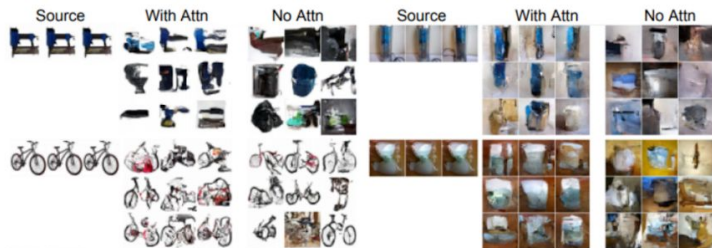
few-shot segmentation



see, e.g.: Shaban, Bansal, Liu, Essa, Boots. **One-Shot Learning for Semantic Segmentation**.
Rakelly, Shelhamer, Darrell, Efros, Levine. **Few-Shot Segmentation Propagation with Guided Networks**.
Dong, Xing. **Few-Shot Semantic Segmentation with Prototype Learning**.

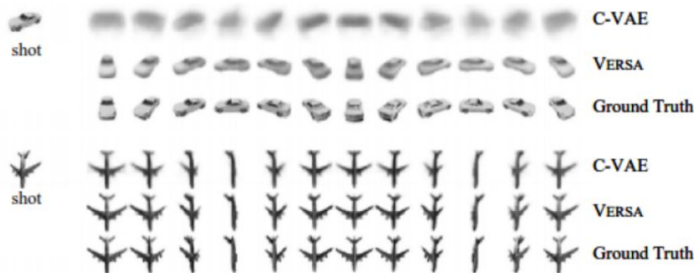
Aplicaciones en generación de imágenes & video

few-shot image generation



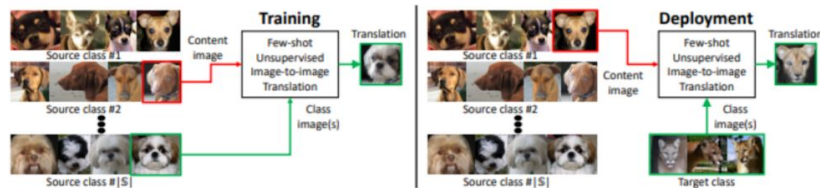
see, e.g.: Reed, Chen, Paine, van den Oord, Eslami, Rezende, Vinyals, de Freitas. **Few-Shot Autoregressive Density Estimation**. and many many others.

generation of novel viewpoints



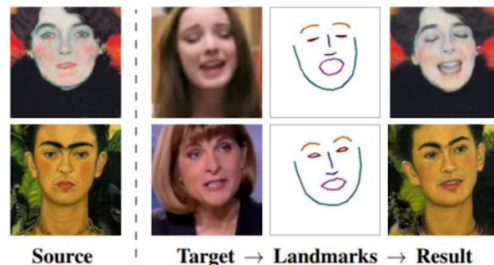
see, e.g.: Gordon, Bronskill, Bauer, Nowozin, Turner. **VERSA: Versatile and Efficient Few-Shot Learning**.

few-shot image-to-image translation



see, e.g.: Liu, Huang, Mallaya, Karras, Aila, Lehtinen, Kautz. **Few-Shot Unsupervised Image-to-Image Translation**.

generating talking heads from images



see, e.g.: Zakharov, Shysheya, Burkov, Lempitsky. **Few-Shot Adversarial Learning of Realistic Neural Talking Head Models**

Aplicaciones en NLP

Adapting to *new programs*

Meta Program Induction

Learn new program from a few I/O examples.

Devlin, Bunel* et al. NeurIPS '17*

Program Synthesis

Question:

How many CFL teams are from York College?

SQL:

```
SELECT COUNT CFL Team FROM  
CFLDraft WHERE College = "York"
```

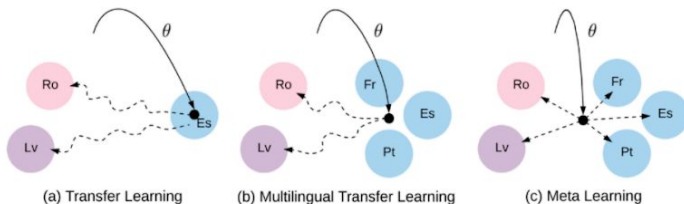
Result:

2

Construct pseudo-tasks with relevance function
Huang et al. NAACL '18

Adapting to *new languages*

Low-Resource Neural Machine Translation



Learn to translate new language pair
w/o a lot of paired data?

Gu et al. EMNLP '18

Learning *new words*

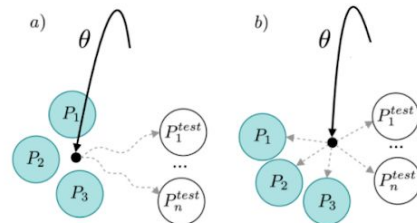
One-Shot Language Modeling

Learn how to use a new word
from one example usage.

Vinyals et al. Matching Networks, '16

Adapting to *new personas*

Personalizing Dialogue Agents



Adapt dialogue to a persona
with a few examples

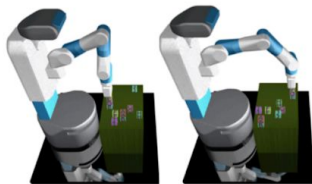
Lin, Madotto* et al. ACL '19*

Aplicaciones en Robótica

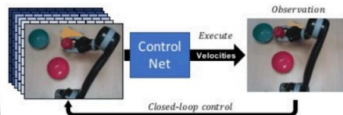
One-Shot *Imitation* Learning

Goal: Given one demonstration of a new task, learn a policy
meta-learning with supervised imitation learning

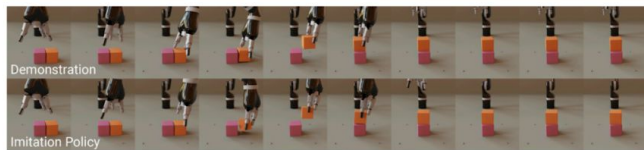
Black-box amortized inference



Duan et al. One-Shot
Imitation Learning '17



James et al. Task-
Embedded Control '18



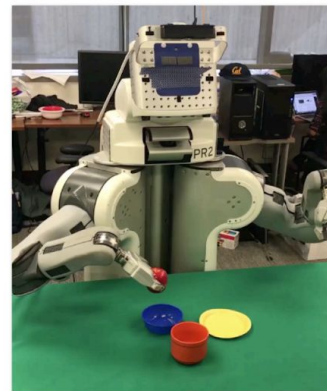
Le Paine et al. One-Shot High Fidelity Imitation '19

Optimization-based inference

Finn*, Yu* et al. Meta Imitation Learning '17

input demo
(via teleoperation)

resulting policy



Also: **One-shot inverse RL** (Xu et al. MandRIL '18, Gleave & Habryka '18), **One-shot hierarchical imitation** (Yu et al. '18)

Aplicaciones en AutoML

- Meta-Learning in HPO & NAS
- Performance Prediction
- Algorithm Selection
- Algorithm Configuration
- Dynamic Algorithm Configuration (DAC)
- Learning to Learn (L2L)

Detalles en: <https://www.automl.org/meta-learning/>

Agenda de la Clase

1. Meta-aprendizaje

- a. Introducción simple
- b. Formalizando algunos conceptos
- c. Relación con Few-shot learning

2. Algunos algoritmos emblemáticos de meta-aprendizaje

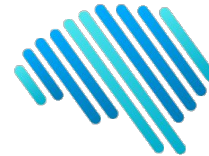
- a. Model-Agnostic Meta-Learning (MAML)
- b. One-shot Learning with Memory-Augmented Neural Networks (MANN)
- c. Métodos no-paramétricos

3. Aplicaciones

4. Trabajo práctico



PONTIFICIA
UNIVERSIDAD
CATÓLICA
DE CHILE



IA Lab

Meta-aprendizaje (meta-learning)

Pablo Messina

Diplomado Inteligencia Artificial

Pontificia Universidad Católica de Chile

Tópicos de Profundización